

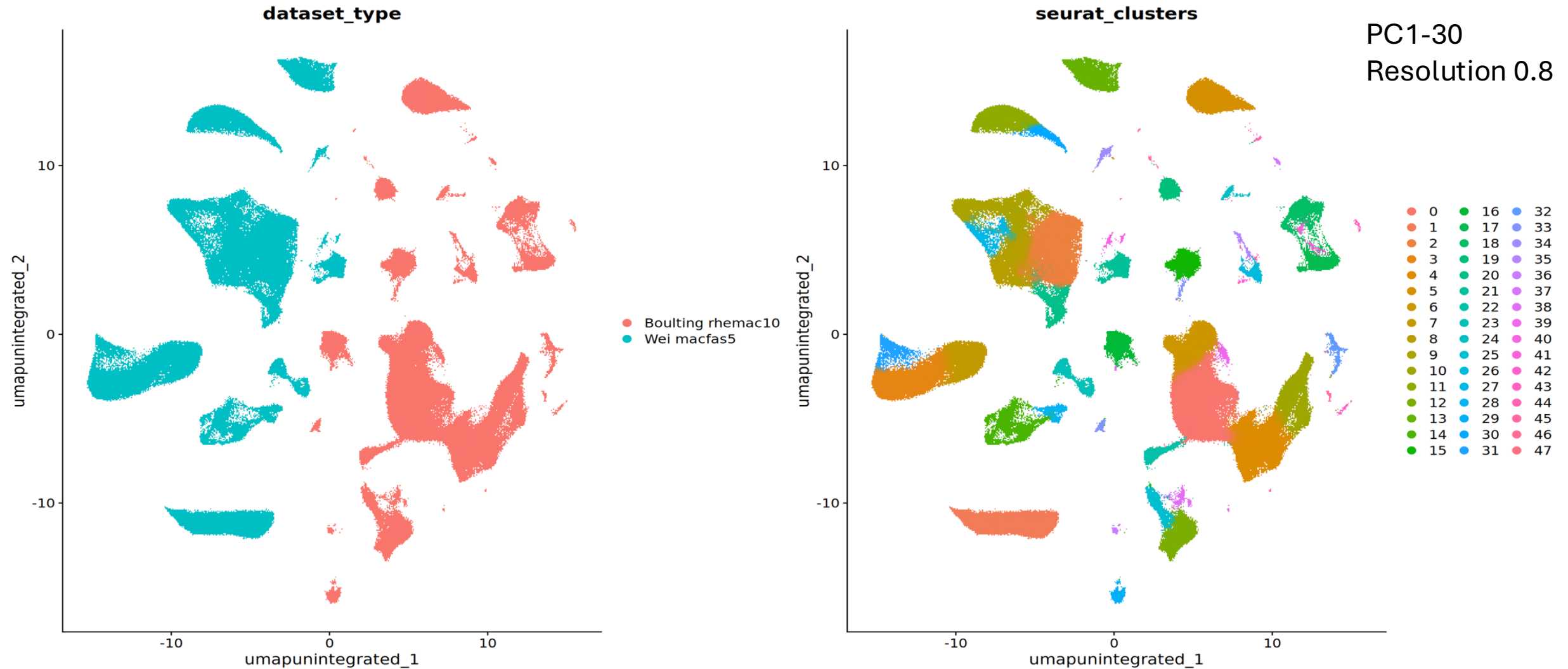
Bioinformatics update

Genevieve Baddoo

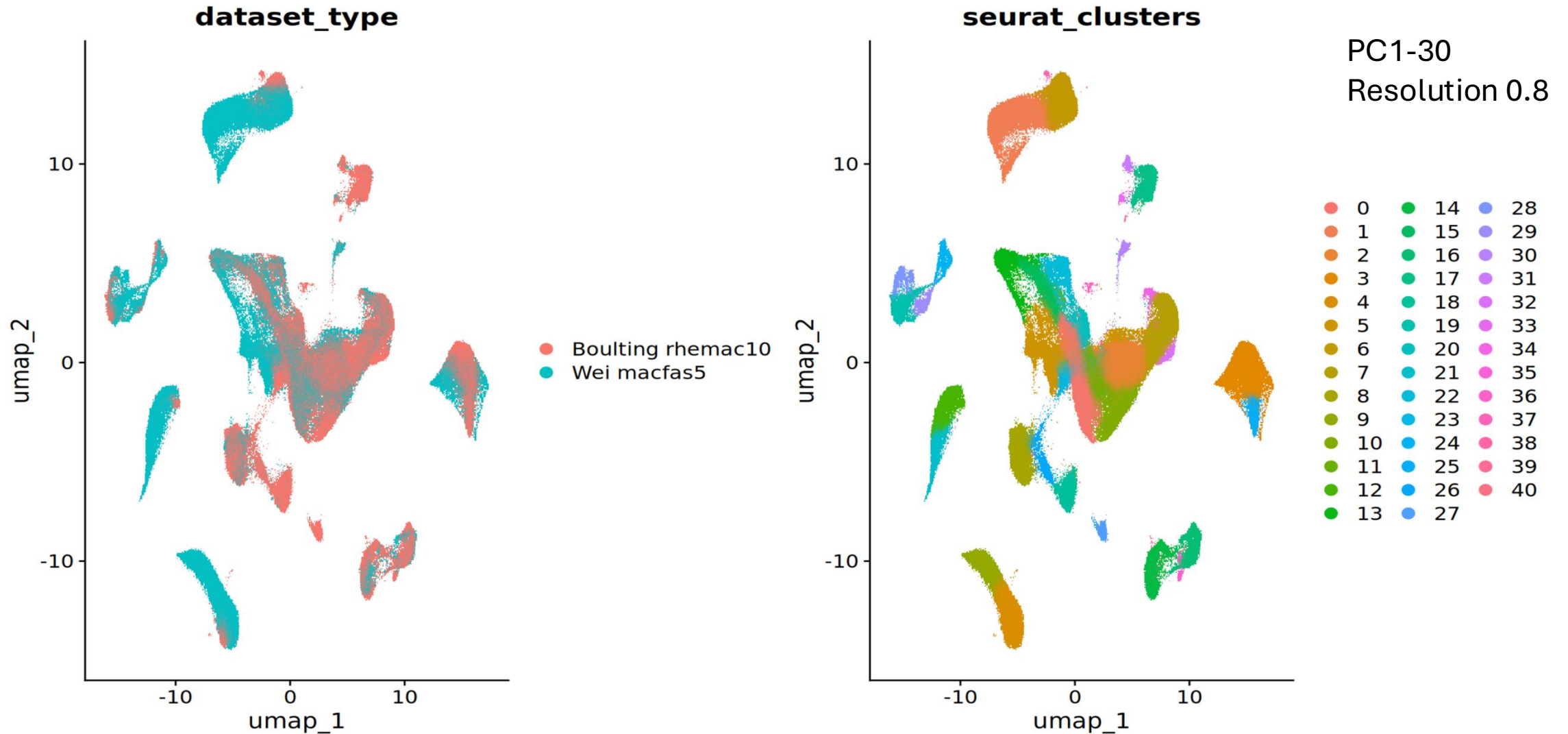
snRNAseq Integration

- Followed [v5.2.0 Seurat tutorial](#)
- Merge Wei et al 2022 & our dataset Seurat objects – counts layer only
- Perform analysis on merged Seurat object without integration
 - `NormalizeData()`, `FindVariableFeatures()`, `ScaleData()`, `RunPCA()`, `FindNeighbors()`, `FindClusters()`, `RunUMAP(merged_obj, method = CCAIntegration, orig.reduction = “pca”, reduction.name = "umap.unintegrated")`
- Perform integration
 - `IntegrateLayers(merged_obj , reduction.name = "integrated.cca")`
 - `JoinLayers()`
 - `FindNeighbors()`, `FindClusters()`, `RunUMAP()`

Boulting & Wei clusters without Integration



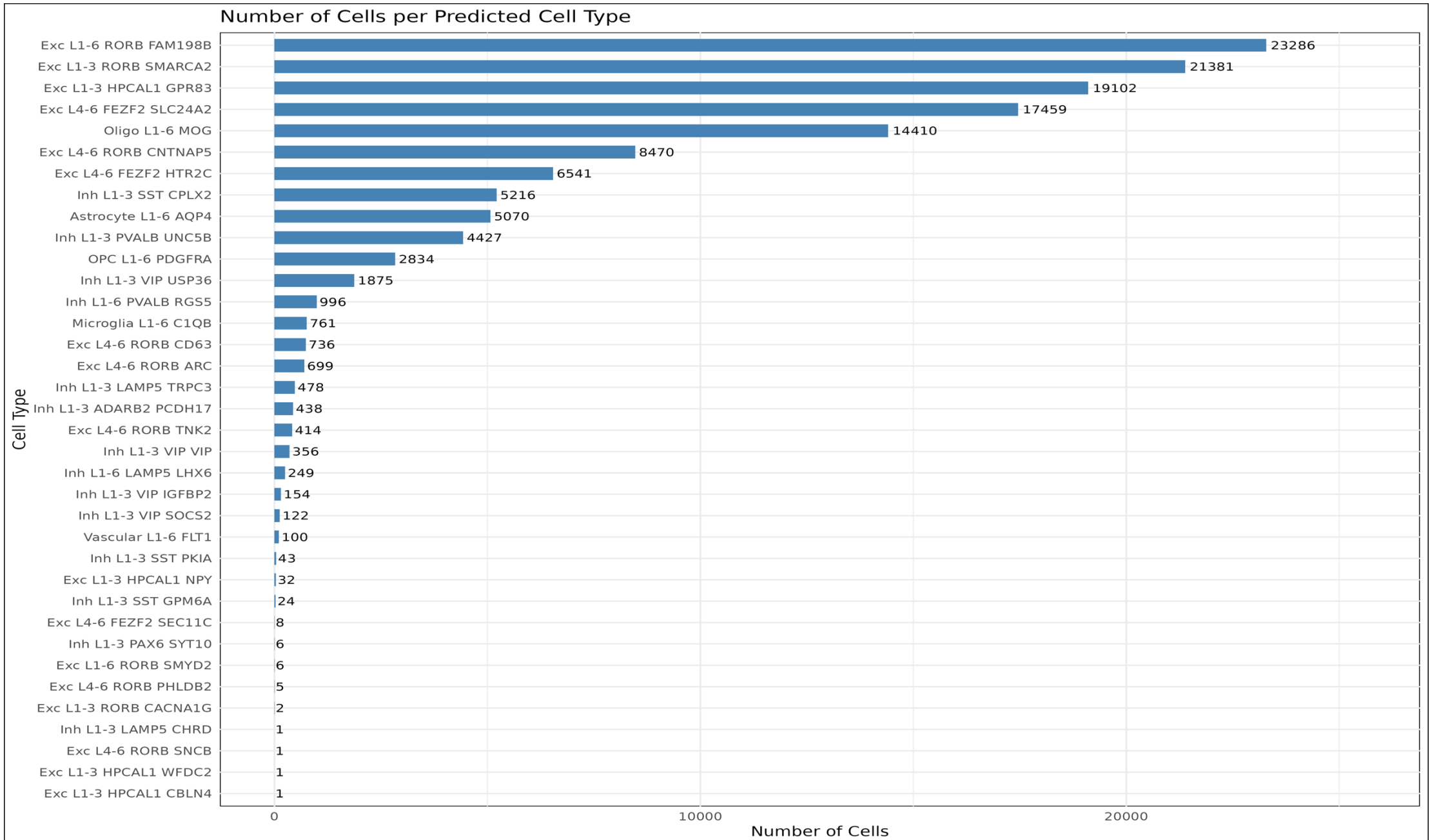
Boulting & Wei clusters after Integration



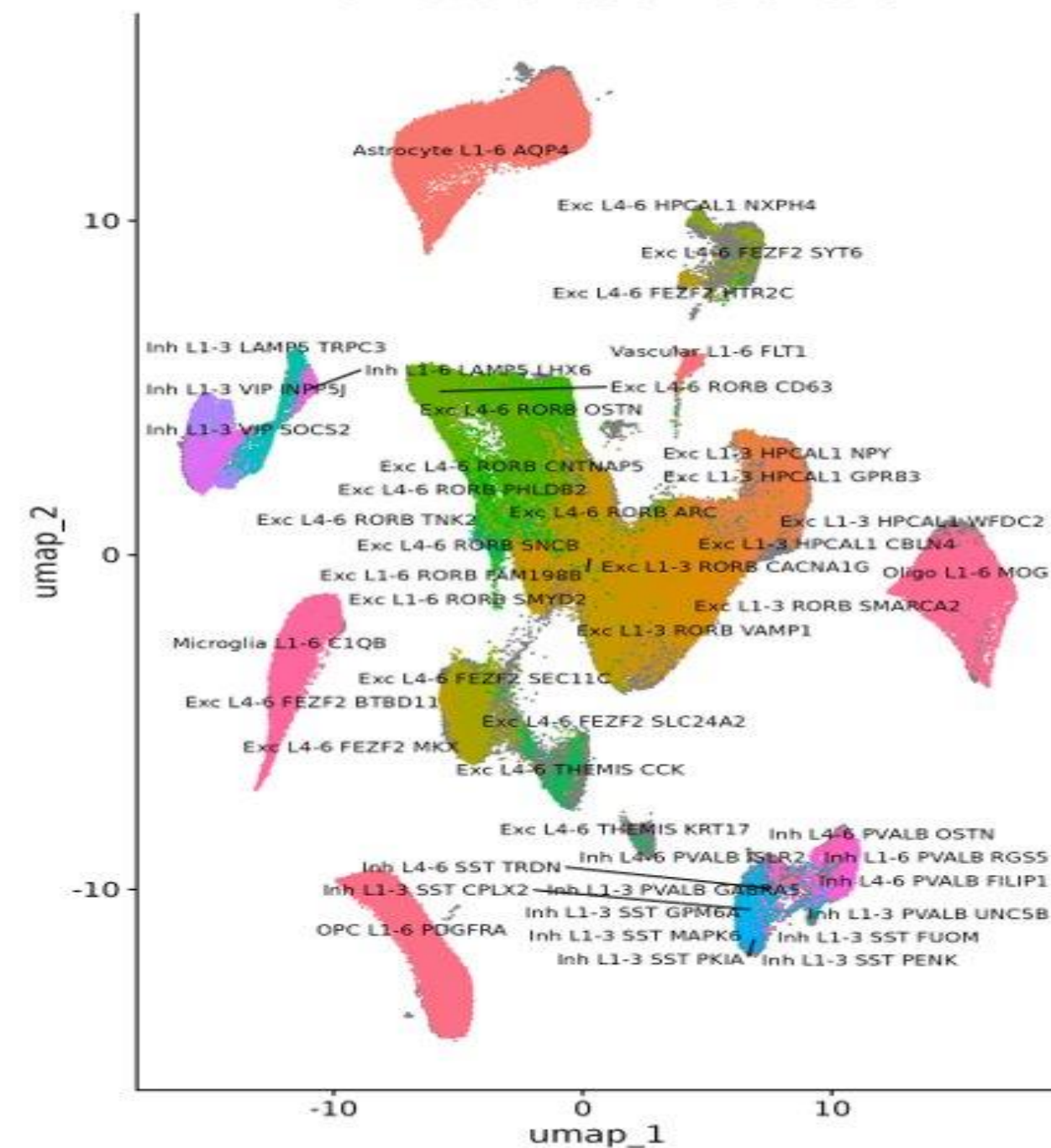
SnRNAseq reference mapping

- Seurat allows the projection of reference data (or meta data) onto a query object.
- Followed [Single-cell Reference Mapping Tutorial](#)
- Mapped and annotated query dataset (our dataset) using Wei et al 2022 dataset
 - FindTransferAnchors()
 - `macaque.anchors <- FindTransferAnchors(reference = wei_macfas5_snRNAseq_so, query = all_combined, dims = 1:30, reference.reduction = "pca")`
 - TransferData()
 - `predictions <- TransferData(anchorset = macaque.anchors, refdata = wei_macfas5_snRNAseq_so$cell_label, dims = 1:30)`
 - AddMetaData()
 - `all_combined <- AddMetaData(all_combined, metadata = predictions)`

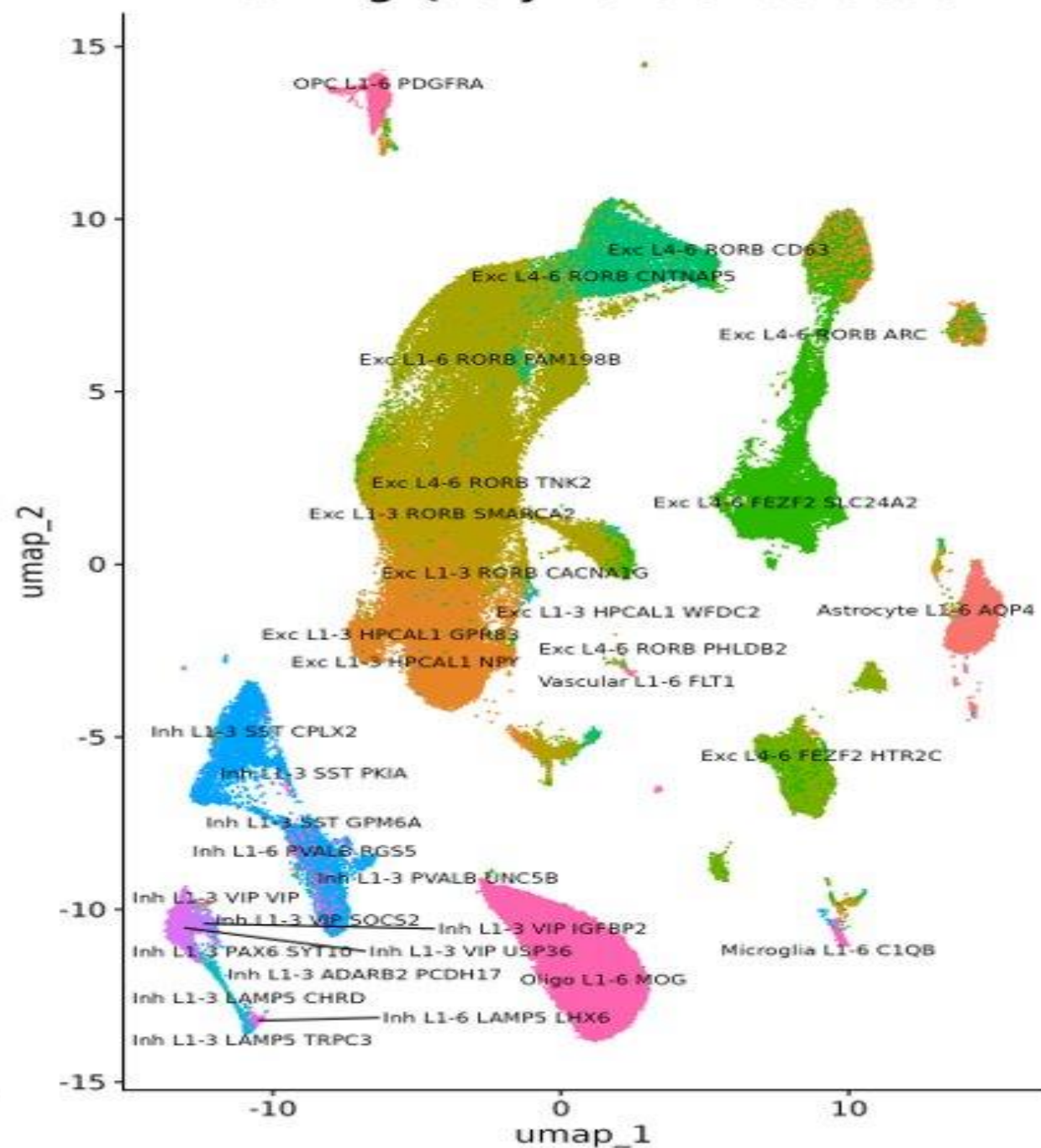
Predicted Cell Types (Wei et al., 2022 Mapping)



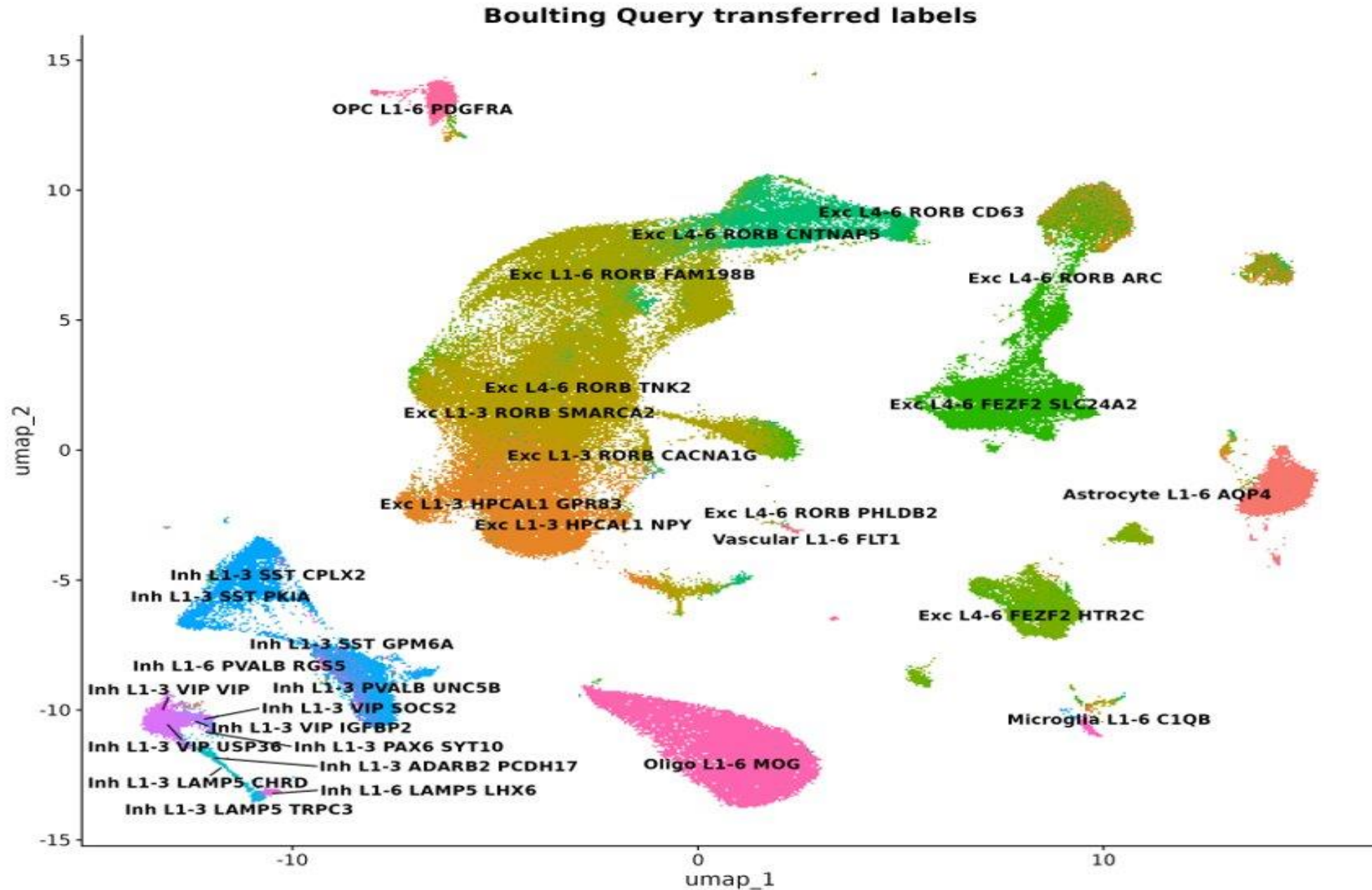
Wei Reference annotations



Boulting Query transferred labels

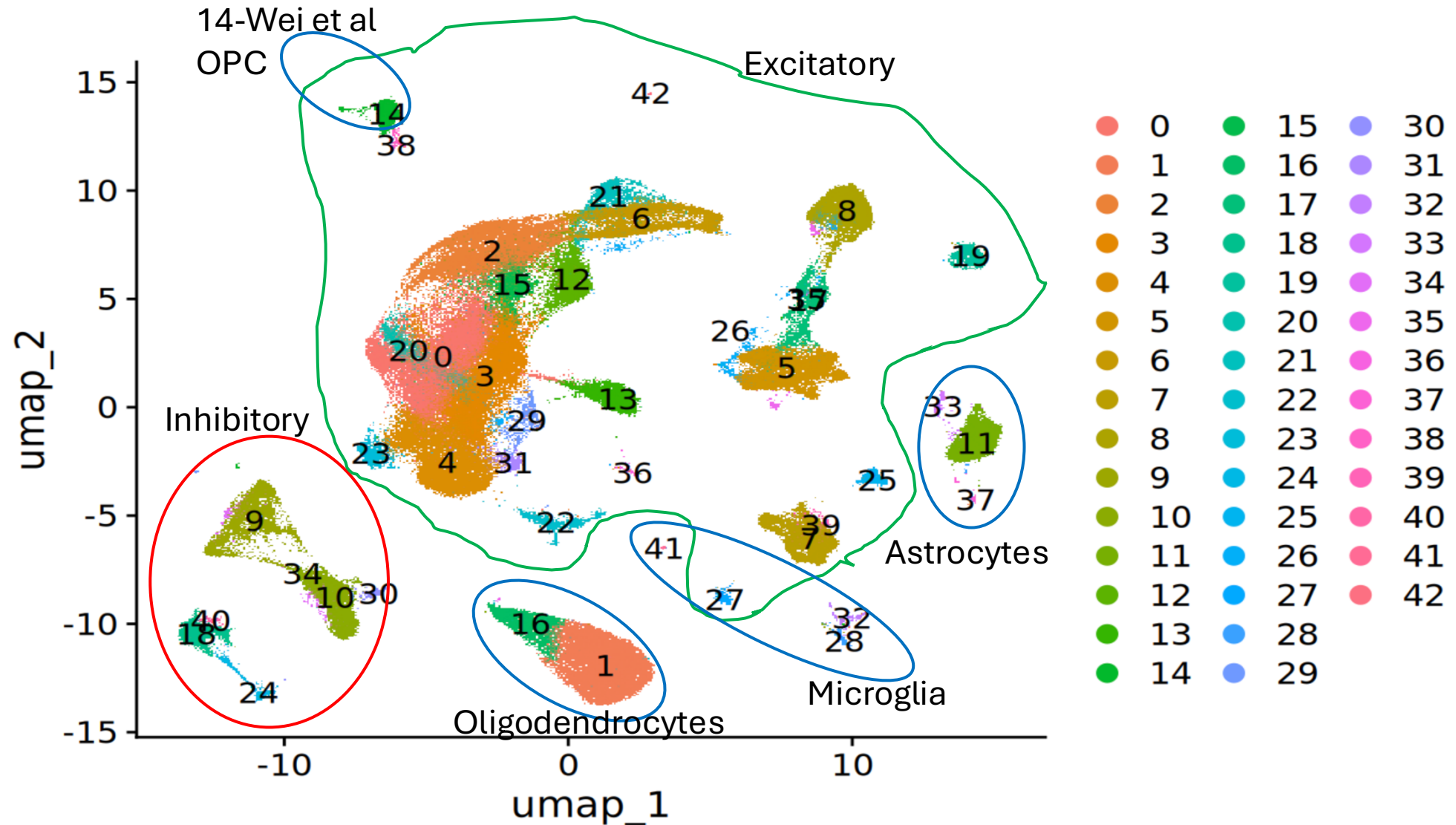


Query transferred labels UMAP

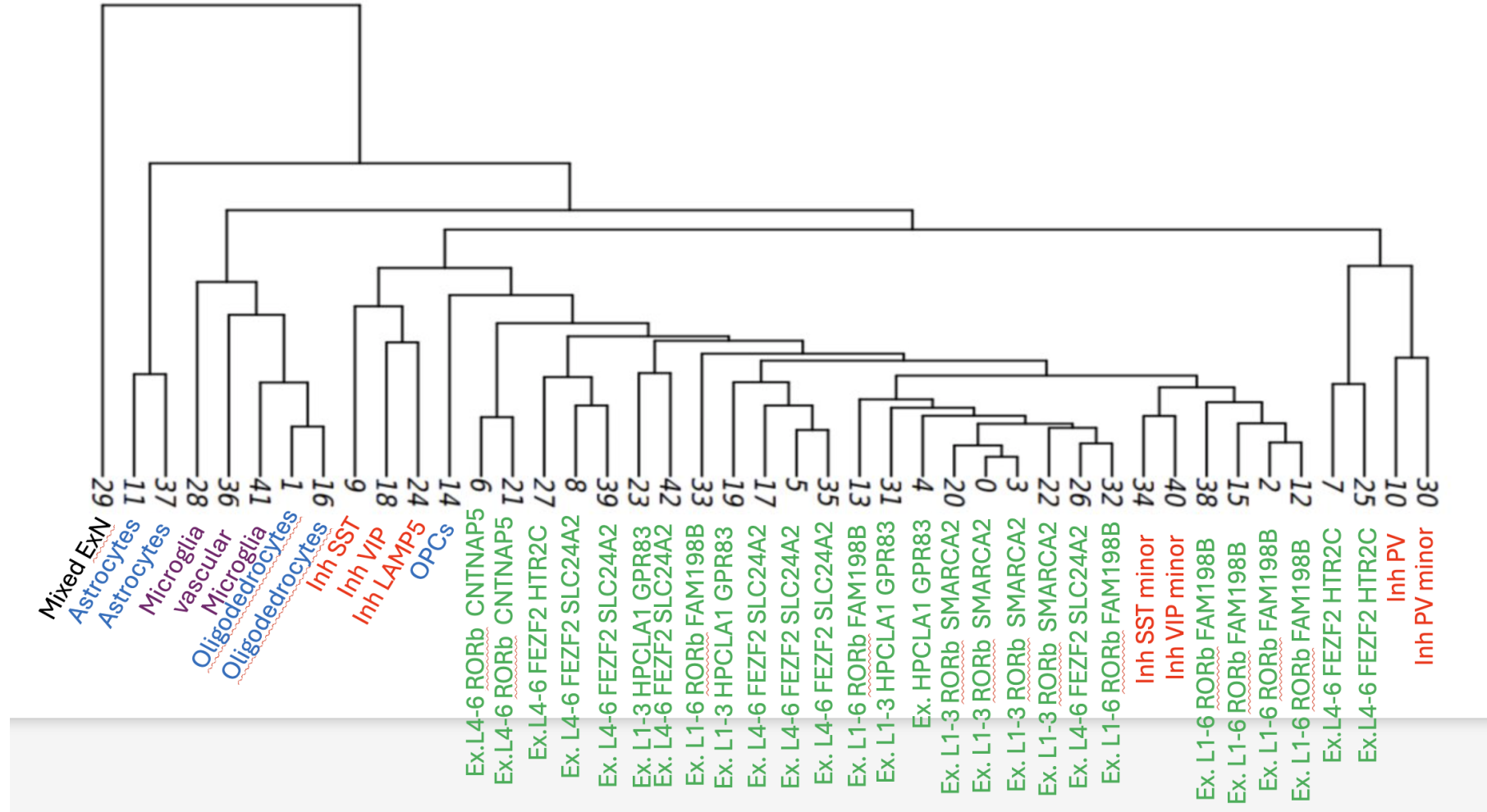


General Categories UMAP

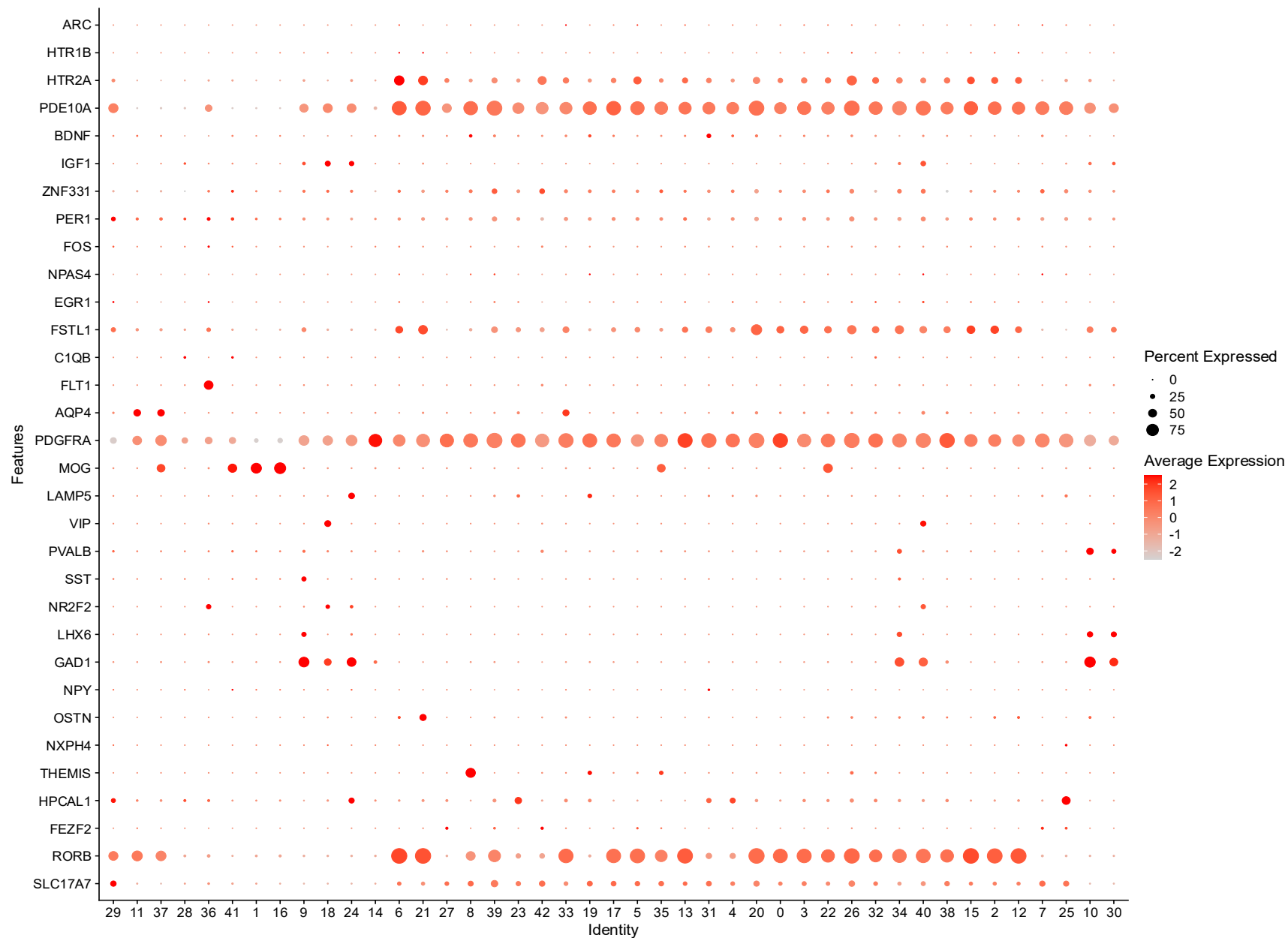
PC1-30
Resolution 0.8



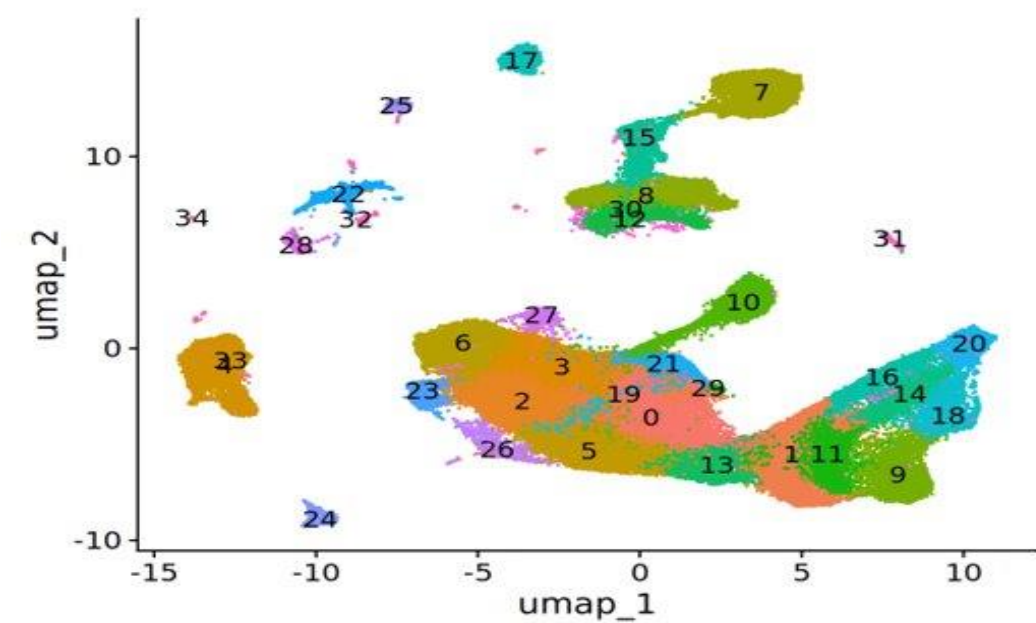
Cell Cluster Relationship Dendrogram



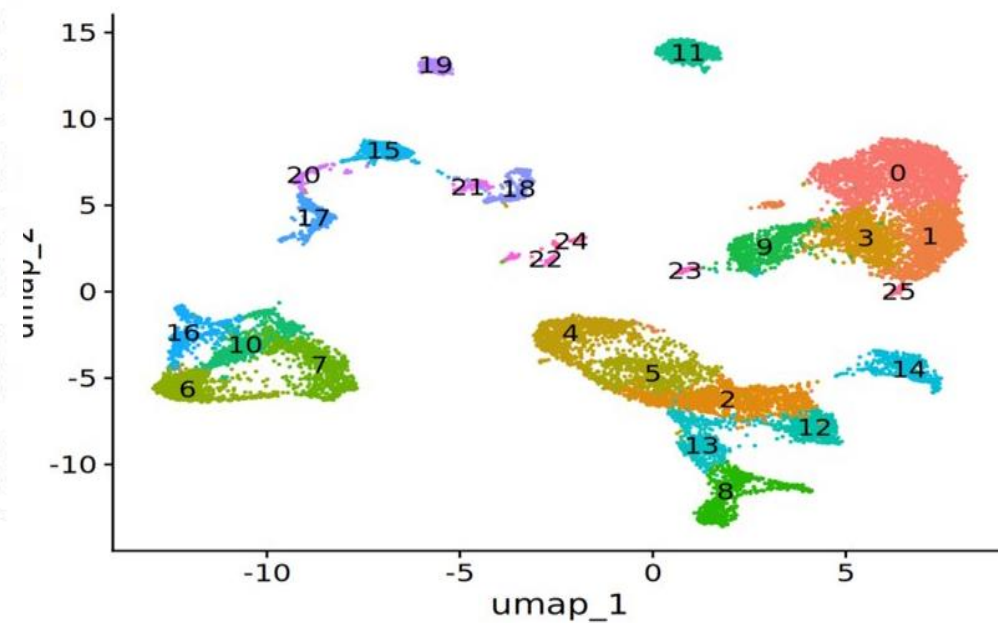
Gene Expression Dotplot



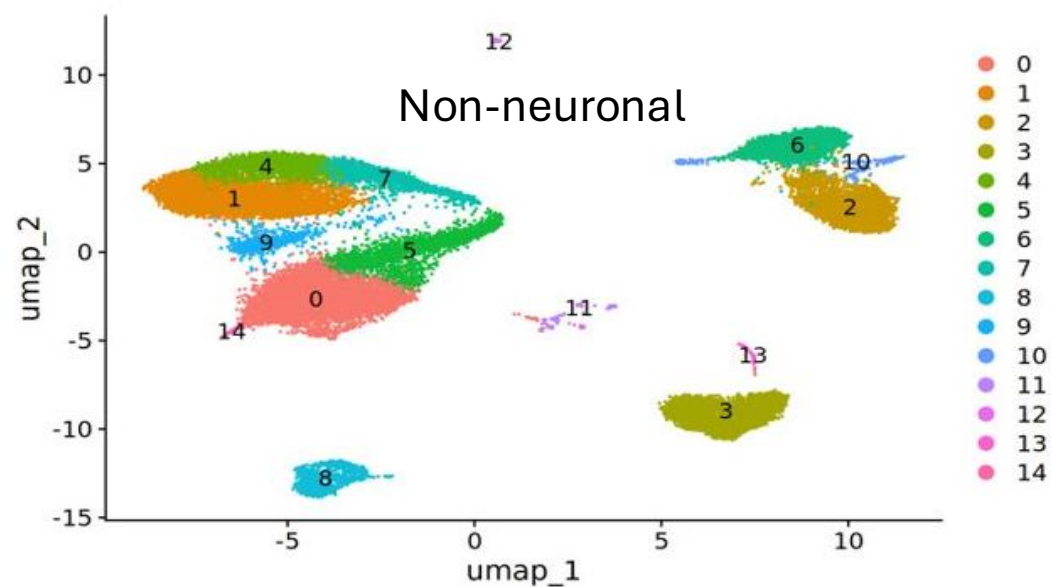
Sub-clustering



Excitatory



Inhibitory



Non-neuronal

Future directions

- Resolve cell type assignments
- Separate activation states within cell types
- Identify new activity-dependent genes through gene-correlation analysis

Thanks!

